

Motor Datasheet: 2026-08-19 17-31-57 output

Machine Configuration

Parameter	Value	Unit
Machine Outer Diameter	34.1	mm
Machine Length / Thickness	17.0	mm
Voltage	48.0	V
Operating Temperature	-40 to 85	°C

Mass Properties

Parameter	Value	Unit
Machine Mass (active components only)	21.6	g
Rotor Mass	8.3	g
Stator Mass	13.4	g
Rotor Structure Included	No	-
Rotor Moment of Inertia	16.8	g·cm ²

Performance Metrics

Parameter	Value	Unit
Torque (continuous)	30.7	mNm
Speed	24500	rpm
Losses (continuous)	13.8	W
Supply Current (RMS)	1.80	A
Phase Resistance (per phase)	1063.2	mΩ
Terminal Inductance ($L_{tt} = L_d + L_q$)	980.2	μH
D-axis Inductance (L_d)	513.9	μH
K_e (peak, phase)	0.93	mV/rpm
K_e (peak, line-to-line)	1.61	mV/rpm
K_t (peak)	13.8	mNm/A
Motor Constant (K_m)	9.6	mNm/√W